BME lecture final term

Textbooks:

[1] R.S. Khurmi & J.K.Gupta, “A Text Book of Thermal Engineering”, two colour edition 1978(Reprint 2003), S.Chand & Company Ltd.

[2] R.S. Khurmi, “A Text Book of Hydraulics, Fluid Mechanics and Hydraulic Machines”, S.Chand & Company Ltd, Reprint 2007

References:

[1] V.P.Vasandani & D.S. Kumar, “Heat Engineering”, Metropolitan Book Co.(Pvt.) Ltd. Reprint 2001

[2] M.J. Moran & H.N. Shapiro, “Engineering Thermodynamics”, John Wiley and Sons.2000

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1.1. Definition

The field of science, which deals with the energies possessed by gases and vapours, is known as *Thermodynamics*. It also includes the conversion of these energies in terms of heat and mechanical work and their relationship with properties of the system. A machine, which converts heat into mechanical work or vice versa, is known as *Heat Engine*. The field of engineering science, which deals with the applications of thermodynamics and its laws to work producing and work absorbing devices, in order to understand their functions and improve their performance, is known as *Thermal Engineering*.

The heat is, usually, generated by the combustion of fuel which may be solid, liquid or gas. It is supplied to the working substance (a source of conveying heat to the heat engine for doing work in the engine cylinder) at a higher temperature. A part of the heat energy is converted into mechanical work by expanding the working substance, within the engine cylinder. The remaining heat energy is rejected at a lower temperature.

The working substances, widely used in the heat engines, are fluids in the gaseous or liquid state. A mixture of air and fuel is used as a working substance in internal combustion engines, and water vapour (steam) in the steam engines or steam turbines.

1.2. Thermodynamic Systems

The *thermodynamic system* (or simply known as *system*) may be broadly defined as a *definite area* or *a space* where some thermodynamic process* is taking place. It is a region where our attention is focussed for studying a thermodynamic process. A little observation will show that a thermodynamic system has its boundaries and anything outside the boundaries is called its *surroundings* as shown in Fig. 1.1. These boundaries may be *fixed* like that of a tank enclosing a certain mass of compressed gas, or *movable* like boundary of a certain volume of liquid in a pipe line.

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Internal energy: dU=dQ-dW

Processes:

1) Adiabatic (entropy constant/no change in heat transfer):

Delta Q=0, So delta U=W, If delta U=+ve, then the system does work, If delta U=-ve, then the work is done on the system.

2) **Isothermal** (constant temperature): delta T=0, So, delta U=0, Because, delta U=TdS-pdV, Or, Q=W, Any energy entering the system must leave as work.

3) **Isobaric** (constant pressure): delta P=0, delta U, Q, W not equal to 0.

4) **Isochoric process** (constant volume): delta volume=0, So, W=0, delta U=delta Q.

1st law, 2nd law, 3rd law and zeroth law of thermodynamics. Limitation of each laws, Heat engine, Kelvin-plank statement, entropy, coefficient of performance (C.O.P), Clausius Theorem of Inequality.

Law of Conservation of Energy

It states. *"The energy can neither be created nor destroyed, though it can be transformed from one form to any other form, in which the energy can exist."*

Laws of Thermodynamics

The following three laws of thermodynamics are important from the subject point of view :

1. Zeroth law of thermodynamics, 2. First law of thermodynamics, and 3. Second law of thermodynamics.

1.49. Zeroth Law of Thermodynamics

This law states, *"When two systems are each in thermal equilibrium with a third system, then the two systems are also in thermal equilibrium with one another."*

This law provides the basis of temperature measurement.

1.50. First Law of Thermodynamics

This law may be stated as follows :

(a) "The heat and mechanical work are mutually convertible". According to this law, when a closed system undergoes a thermodynamic cycle, the net heat transfer is equal to the net work transfer. In other words, the cyclic integral of heat transfers is equal to the cyclic integral of work transfers. Mathematically,

$$\oint \delta Q = \oint \delta W$$

where symbol $\oint$ stands for cyclic integral (integral around a complete cycle), and δQ and δW represent infinitesimal elements of heat and work transfers respectively. It may be noted that δQ and δW are expressed in same units.

(b) The energy can neither be created nor destroyed though it can be transformed from one form to another. According to this law, when a system undergoes a change of state (or a thermodynamic process), then both heat transfer and work transfer takes place. The net energy transfer is stored within the system and is known as stored energy or total energy of the system. Mathematically

$$\delta Q - \delta W = dE$$

The symbol δ is used for a quantity which is inexact differential and symbol d is used for a quantity which is an exact differential. The quantity E is an extensive property and represents the total energy of the system at a particular state.

On integrating the above expression for a change of state from 1 to 2, we have

$$Q_{1-2} - W_{1-2} = E_2 - E_1 \qquad \ldots (Q, W \text{ and } E \text{ are in same units})$$

For a unit mass, this expression is written as

$$q_{1-2} - w_{1-2} = e_2 - e_1$$

Q_{1-2} = Heat transferred to the system during the process from state 1 to state 2,

W_{1-2} = Workdone by the system on the surroundings during the process, and

E_1 = Total energy of the system at state 1

$$= PE_1 + KE_1 + U_1 = *m\,g\,z_1 + \frac{m\,V_1^2}{2} + U_1$$

E_2 = Total energy of the system at state 2

$$= PE_2 + KE_2 + U_2 = m\,g\,z_2 + \frac{m\,V_2^2}{2} + U_2$$

Thus the above expression may be written as

$$Q_{1-2} - W_{1-2} = E_2 - E_1 \quad \ldots (i)$$

$$= (PE_2 + KE_2 + U_2) - (PE_1 + KE_1 + U_1)$$

$$= (PE_2 - PE_1) + (KE_2 - KE_1) + (U_2 - U_1) \quad \ldots (ii)$$

$$= m\,(g\,z_2 - g\,z_1) + m\left(\frac{V_2^2}{2} - \frac{V_1^2}{2}\right) + (U_2 - U_1)$$

For unit mass, this expression is written as

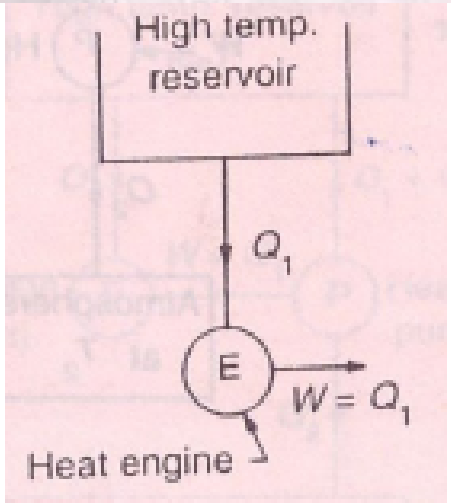
$$q_{1-2} - w_{1-2} = (g\,z_2 - g\,z_1) + \left(\frac{V_2^2}{2} - \frac{V_1^2}{2}\right) + (u_2 - u_1)$$

1.52. Second Law of Thermodynamics

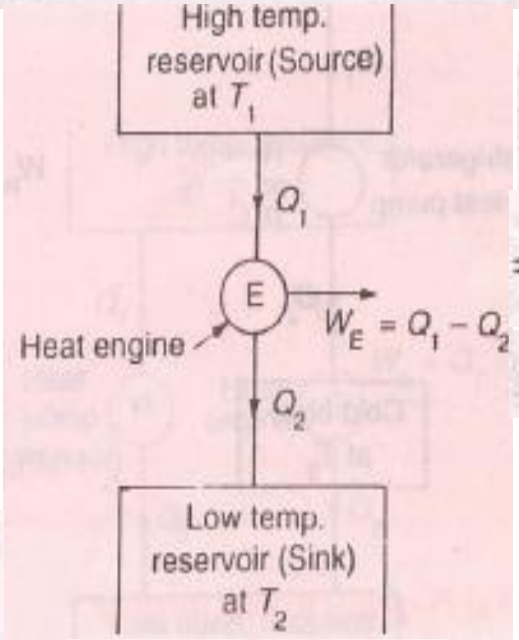
The second law of thermodynamics may be defined in many ways, but the two commo statements according to Kelvin - Planck and Clausius are as follows :

1. *Kelvin - Planck Statement.* According to Kelvin-Planck *'It is impossible to construct an engine working on a cyclic process, whose sole purpose is to convert heat energy from a single thermal reservoir* into an equivalent amount of work'.* In other words, no actual heat engine, working on a cyclic process, can convert whole of the heat supplied to it, into mechanical work. It means that there is a degradation of energy in the process of producing mechanical work from the heat supplied. Thus the Kelvin - Planck statement of the second law of thermodynamics, is sometimes known as *law of degradation of energy.*

A heat engine which violates this statement of the second law of thermodynamics (*i.e.* a heat engine which converts whole of the heat energy into mechanical work) is known as ***perpetual motion machine of the second kind* (briefly written as PMM-II) or 100 percent efficient machine which is impossible to obtain in actual practice, because no machine can convert whole of the heat energy supplied to it, into its equivalent amount of work.



(*a*) Perpetual motion machine of the second kind (impossible).



(*b*) Heat engine.

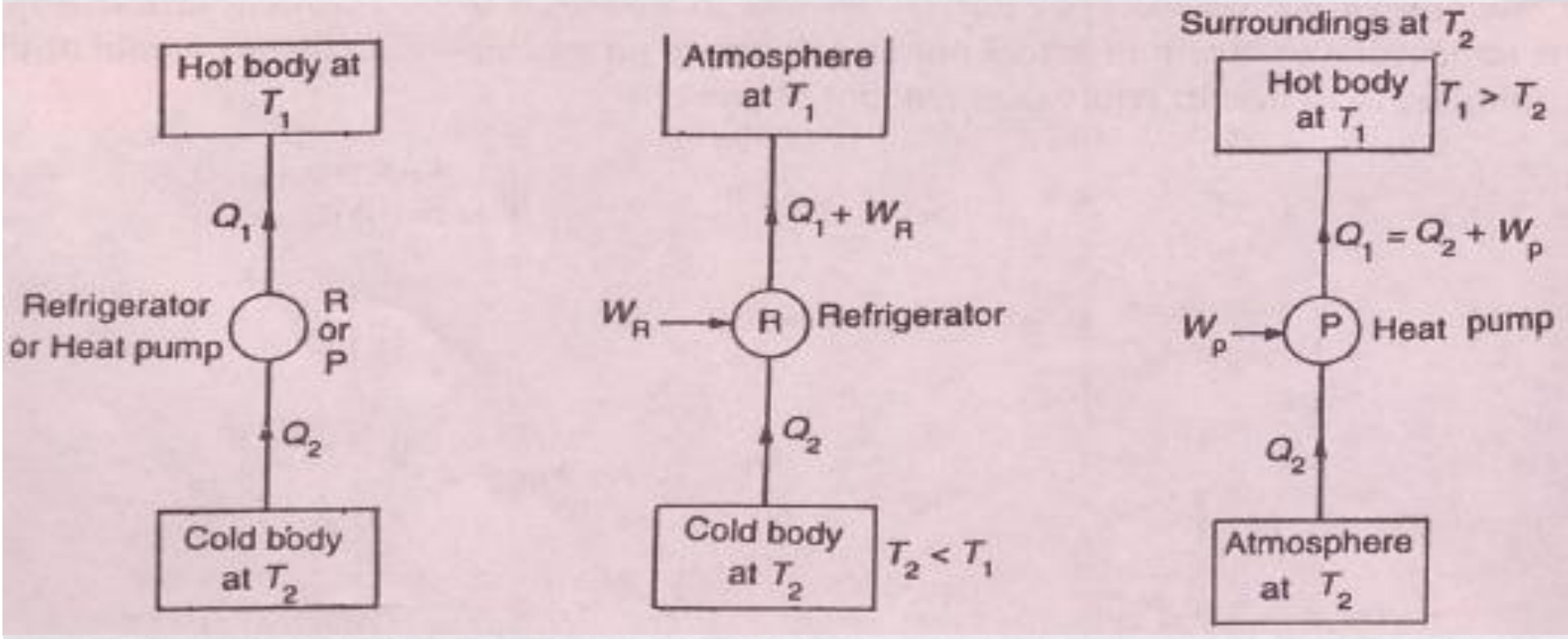
$$\eta_{max} = \frac{\text{Maximum work obtained}}{\text{Total heat supplied}}$$

$$= \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$$

For a reversible engine, $Q_1 / T_1 = Q_2 / T_2$.

2. *Clausius Statement.* According to Clausius statement "*It is impossible for a self acting machine, working in a cyclic process, to transfer heat from a body at a lower temperature to a body at a higher temperature without the aid of an external agency.*" In other words, heat cannot flow itself from a cold body to a hot body without the help of an external agency (*i.e.* without the expenditure of mechanical work).

The device (such as a refrigerator or a heat pump), as shown in Fig. 1.12 (*a*), violates the Clausius statement because no input work is supplied to the device to transfer heat from a cold body to a hot body. Such a device is called perpetual motion machine of the second kind.



(*a*) Perpetual motion machine of the second kind.

(*b*) Refrigerator.

(*c*) Heat pump.

In order to achieve the object of transferring heat from a cold body to a hot body, the refrigerator and a heat pump, while operating in a cyclic process, require an input work, as shown in Fig. 1.12 (*b*) and (*c*) respectively. Though there is no difference between the cycle of operations of the refrigerator and a heat pump and achieve the same overall objective, but the basic purpose of each is quite different. A refrigerator is a device which operating in a cyclic process, maintains the temperature of a cold body (refrigerated space) at a temperature lower than the temperature of the surroundings. On the other hand, a heat pump is a device which operating in a cyclic process, maintains the temperature of a hot body (heated space) at a temperature higher than the temperature of surroundings. In other words, a refrigerator works between the cold body temperature and the *atmospheric temperature whereas a heat pump operates between the hot body temperature and the atmospheric temperature.

The **performance of refrigerator and heat pump is measured in terms of coefficient of performance which is defined as the ratio of the maximum heat transferred (*i.e.* heat taken from the cold body) to the amount of work required to produce the desired effect. Mathematically, maximum coefficient of performance for a refrigerator,

$$(\text{C.O.P})_R = \frac{Q_2}{W_R} = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2}$$

and maximum coefficient of performance for a heat pump,

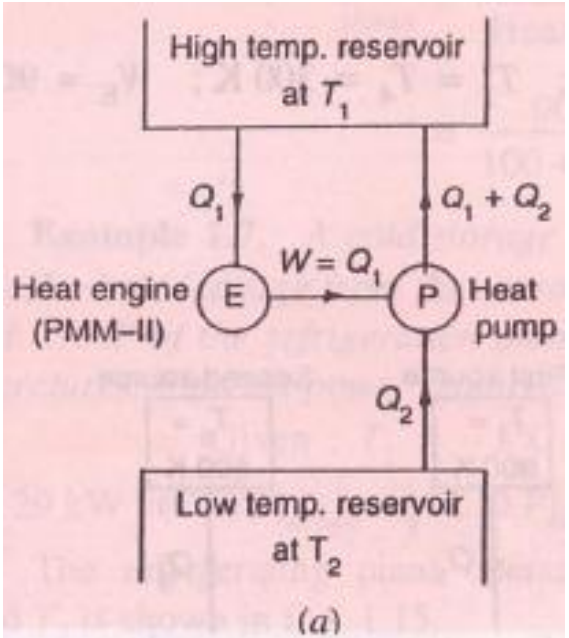
$$(\text{C.O.P})_P = \frac{Q_1}{W_P} = \frac{Q_1}{Q_1 - Q_2} = \frac{T_1}{T_1 - T_2} = \frac{T_2}{T_1 - T_2} + 1$$

$$= (\text{C.O.P})_R + 1$$

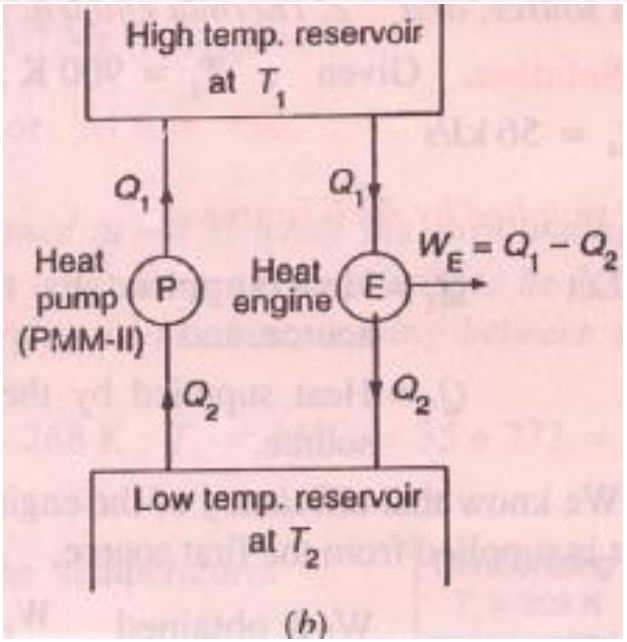
We see that C.O.P of a heat pump is greater than C.O.P of a refrigerator by unity.

1.53. Equivalence of Kelvin-Planck and Clausius Statements

Though Kelvin-Planck and Clausius statements of the second law of thermodynamics appear to be different, from each other, but these two statements are virtually equivalent in all respects. The equivalence of the Kelvin-Planck and Clausius statements can be proved if it can be shown that the violation of Kelvin-Planck statement implies the violation of Clausius statement and vice versa. This is discussed as follows :

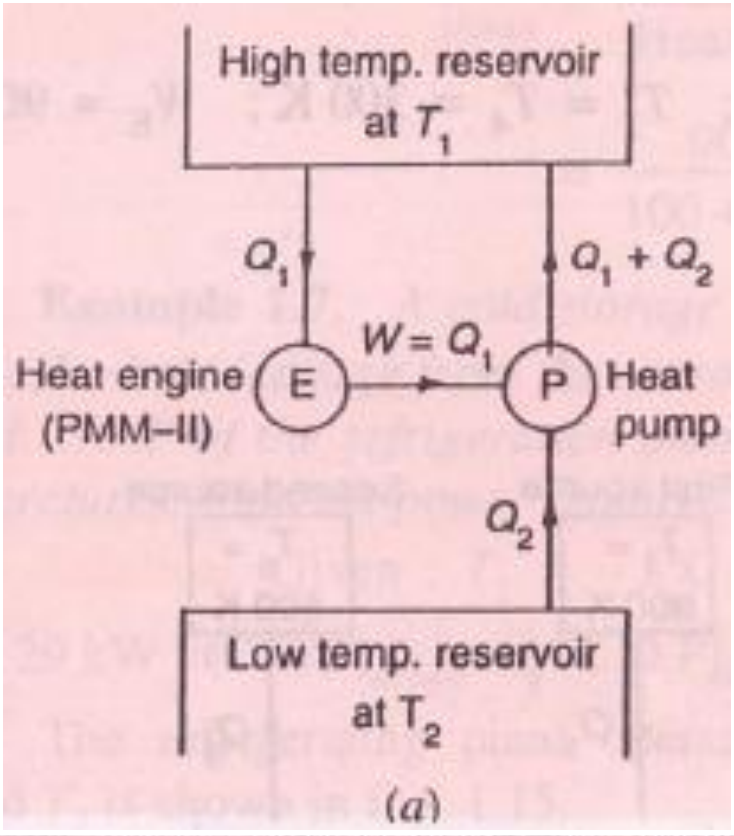


(a)

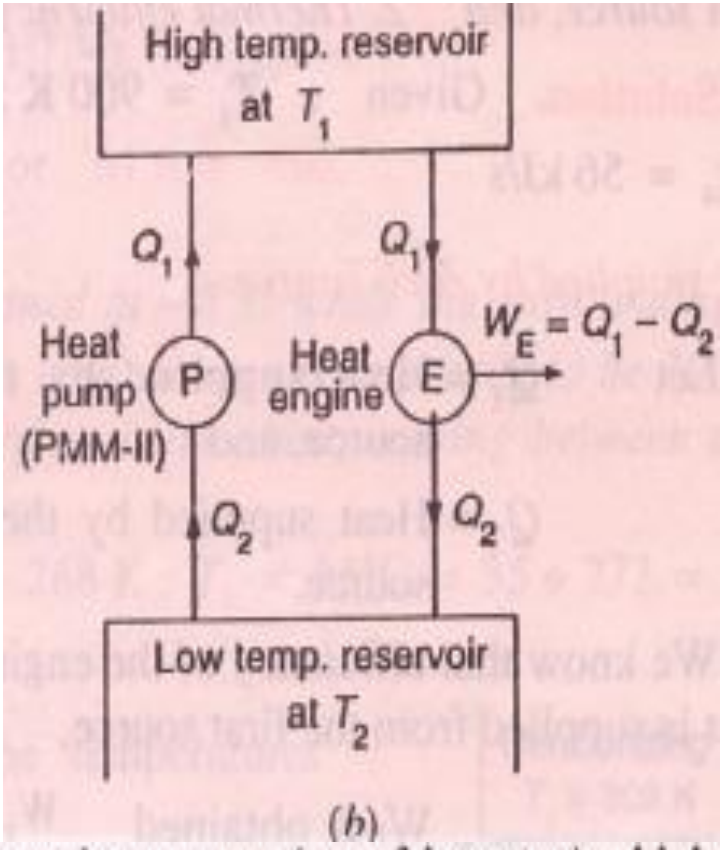


(b)

1. Consider a system as shown in Fig. 1.13 (*a*). In this system, a heat engine having 100 percent thermal efficiency (*i.e.* PMM-II) is violating the Kelvin-Planck statement as it converts the heat energy (Q_1) from a single high temperature reservoir at T_1, into an equivalent amount of work (*i.e.* $W = Q_1$). This work output of the heat engine can be used to drive a heat pump (or refrigerator) which

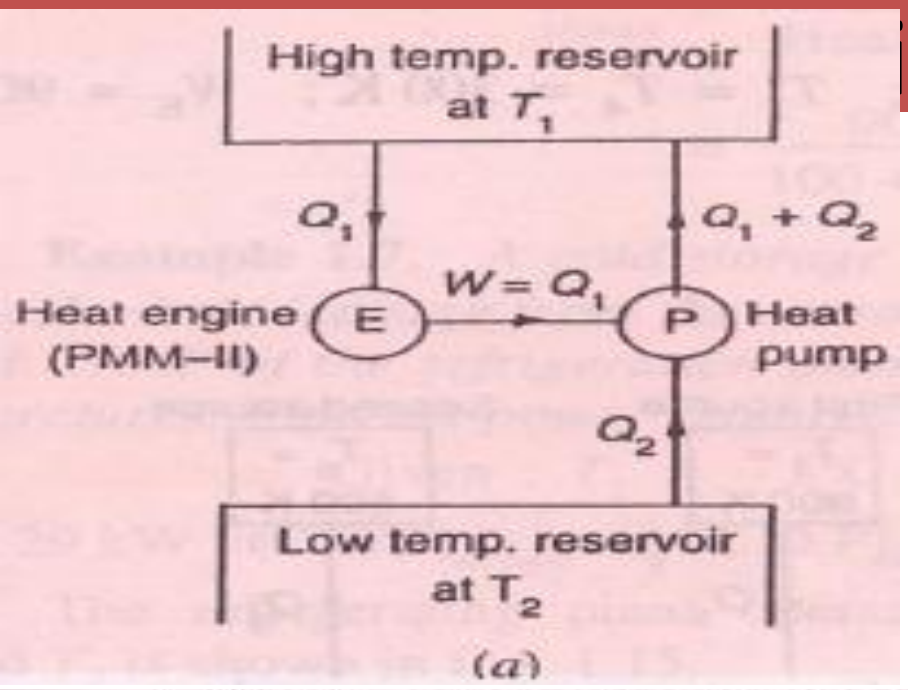


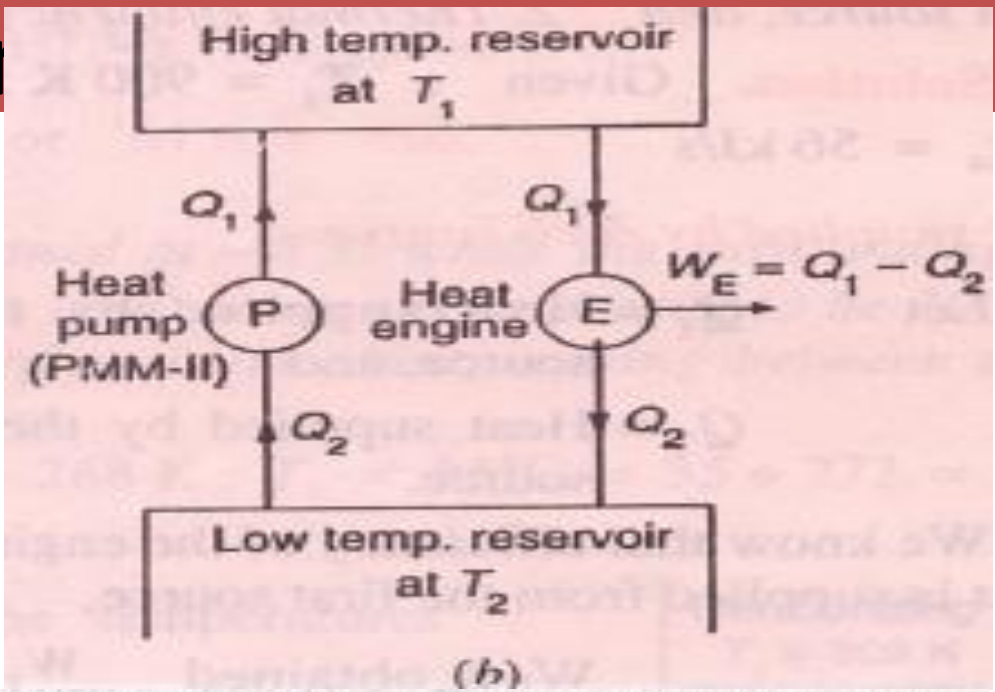
(a)



(b)

$W = Q_1$). This work output of the heat engine can be used to drive a heat pump (or refrigerator) which receives an amount of heat Q_2 from a low temperature reservoir at T_2 and rejects an amount of heat $(Q_1 + Q_2)$ to a high temperature reservoir at T_1. If the combination of a heat engine and a heat pump (or refrigerator) is considered as a single system, as shown in Fig. 1.13 (*a*), then the result is a device that operates in a cycle and has no effect on the surroundings other than the transfer of heat Q_2 from a low temperature reservoir to a high temperature reservoir, thus violating the Clausius statement. Hence, a violation of Kelvin-Planck statement leads to a violation of Clausius statement.





2. Consider a system as shown in Fig. 1.13 (*b*). In this system, a heat pump or refrigerator (*i.e.* PMM-II) is violating the Clausius statement as it transfers heat from a low temperature reservoir at T_2 to a high temperature reservoir at T_1 without any expenditure of work. Now let a heat engine, operating between the same heat reservoirs, receives an amount of heat Q_1 (as discharged by the heat pump) from the high temperature reservoir at T_1, does work ($W_E = Q_1 - Q_2$) and rejects an amount of heat Q_2 to the low temperature reservoir at T_2. If the combination of the heat pump (or refrigerator) and the heat engine is considered as a single system, as shown in Fig. 1.13 (*b*), then the result is a device that operates in a cycle whose sole effect is to remove heat at the rate of ($Q_1 - Q_2$) and convert it completely into an equivalent amount of work, thus violating the Kelvin-Planck statement. Hence, a violation of Clausius statement leads to a violation of Kelvin-Planck statement.

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34.1. Introduction

As a matter of fact, the subject of heat transfer is getting more and more importance in the sphere of science and technology these days. The mechanical engineers deal with the problems of heat transfer in the design of their I.C. engines, refrigeration and air-conditioning system as well as various types of steam generation plants. The electrical engineers deal with the problems of heat transfer in the cooling systems of power generation machines, transformers and various types of huge electrical installations. Similarly, the civil engineers require the knowledge of heat transfer in the design of dams, tunnels, cold storages, cinema halls and other important structures.

The subject of heat transfer is very vast, and its detailed study is beyond the scope of this book. However, its basic principles will be discussed in this book.

34.2. Methods of Heat Transfer

The heat transfer may be broadly defined as the transmission of heat energy from one region to another due to the temperature difference between these two regions.

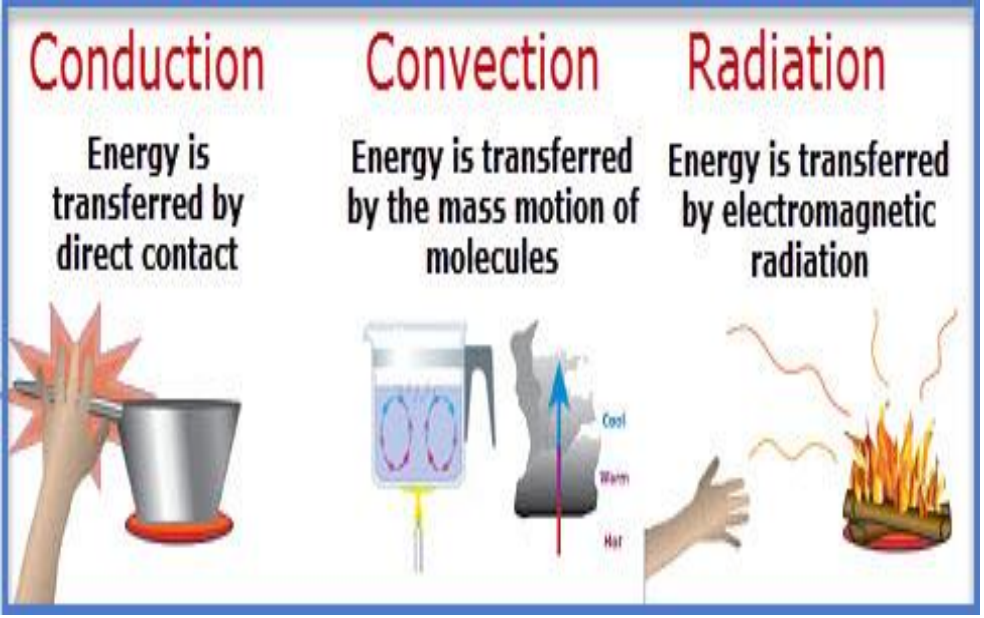
The following methods, of heat transfer, from one body to another, are important from the subject point of view :

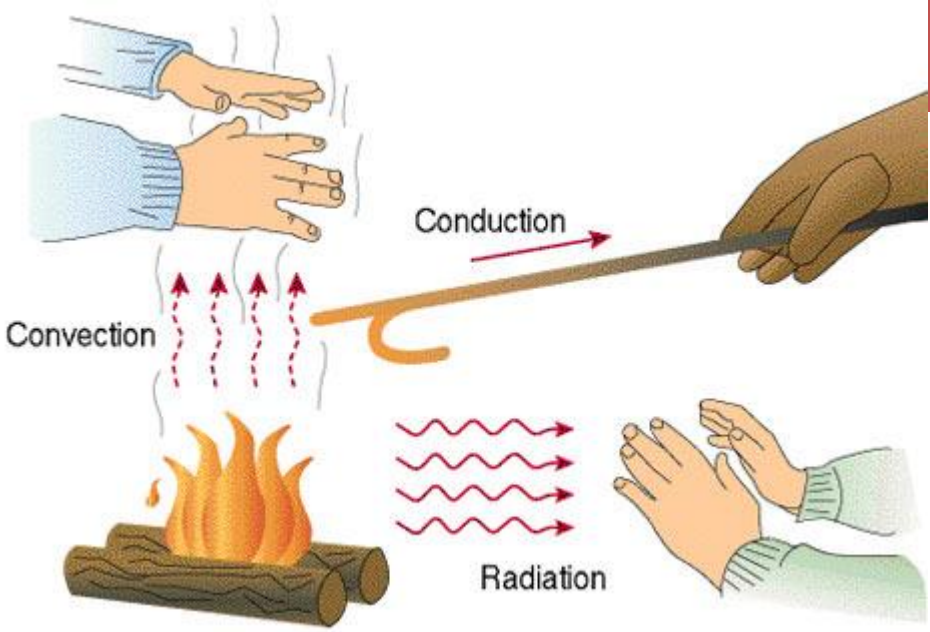
1. *Conduction.* It is a process of heat transfer from one particle of the body to another in the direction of fall of temperature. The particles themselves remain in fixed position relative to each other.

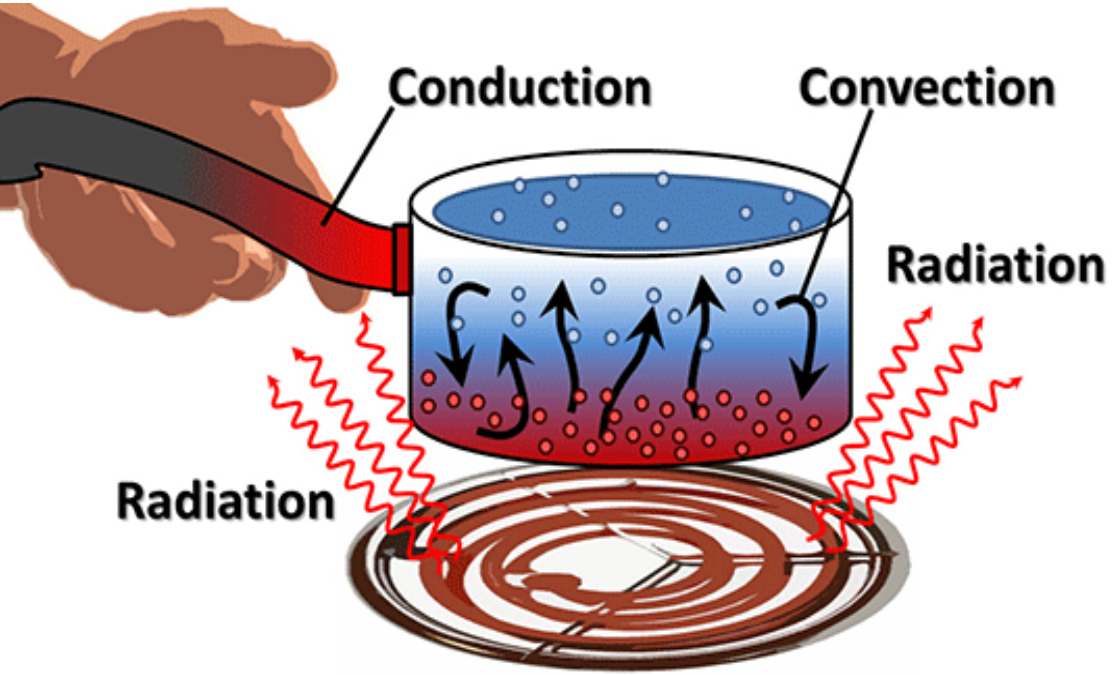
2. *Convection.* It is a process of heat transfer from one particle of the body to another by convection current. In this case, the particles of the body move relative to each other.

3. *Radiation.* It is a process of heat transfer from a hot body to a cold body, in a straight line, without affecting the intervening medium.

In this chapter, we shall discuss the heat transfer by conduction only.







CONVECTION
the transfer of heat through a fluid (liquid or gas) caused by molecular motion

CONDUCTION
the transfer of heat or electric current from one substance to another by direct contact.

RADIATION
energy that is radiated or transmitted in the form of rays or waves or particles

http://www.greenspec.co.uk/building-design/heat-transfer-conduction-convection-radiation/

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34.3. Newton's Law of Cooling

It is an important law in the field of heat transfer which states, "*Heat transfer from a hot body to a cold body is directly proportional to the surface area and difference of temperatures between the two bodies*".

It is a general law, for the heat transfer which can not be applied to all sets of conditions. But it paved the way for other laws dealing in the heat loss.

34.4. Fourier's* Law of Heat Conduction

It is also an important law in heat conduction, which is represented by the equation,

$$Q \propto A \times \frac{dT}{dx} = k A \times \frac{dT}{dx}$$

where

Q = Amount of heat flow through the body in a unit time.

A = Surface area of heat flow. It is taken at right angles to the direction of flow.

dT = Temperature difference on the two faces of the body.

dx = Thickness of the body through which the heat flows. It is taken along the direction of heat flow.

k = Constant of proportionality known as *thermal conductivity* of the body.

The **time rate** of heat transfer through a material is proportional to the negative gradient in the **temperature** and to the **area**, at right angles to that gradient, through which the heat flows.

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The **time rate** of heat transfer through a material is proportional to the negative gradient in the **temperature** and to the **area**, at right angles to that gradient, through which the heat flows.

$$Q \propto A \times \frac{dT}{dx} = kA \times \frac{dT}{dx}$$

Where

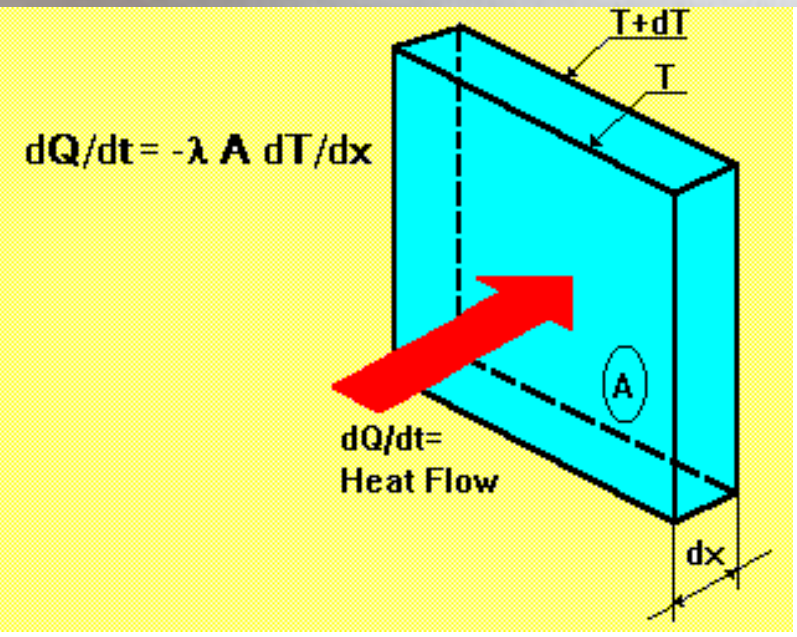
Q = Amount of heat flow through the body in a unit time.

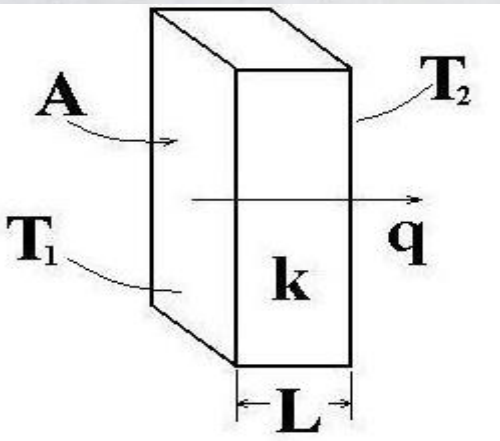
A = Surface area of heat flow. It is taken at right angles to the direction of flow.

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k = Constant of proportionality known as *thermal conductivity* of the body.

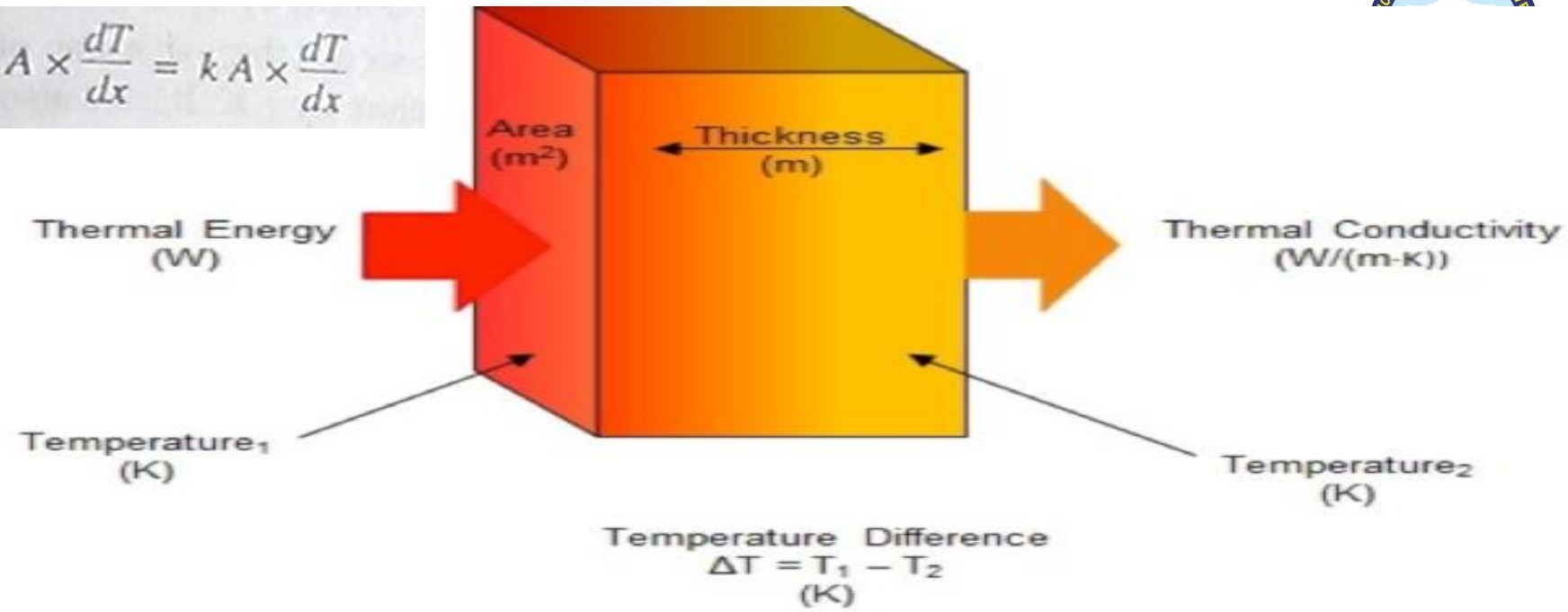


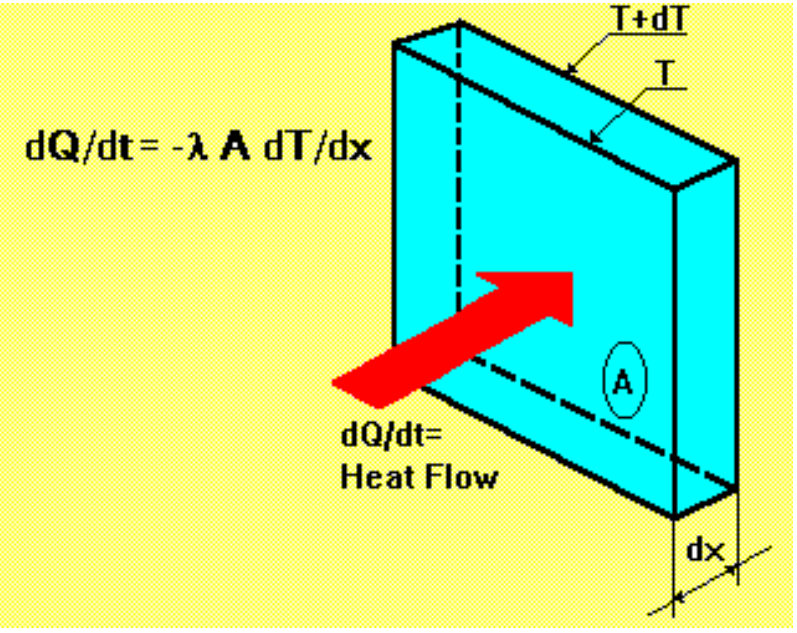


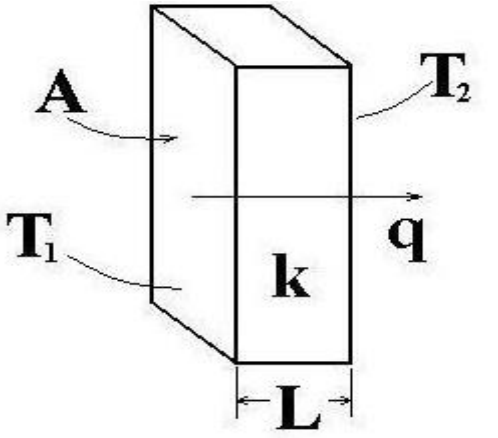
$$q = k A \frac{T_1 - T_2}{L}$$

Thermal conductivity [κ, k or λ measured in W/(m·K)] expresses its ability to conduct heat, a rate measurement of **heat flow**, through a material of a **given thickness**, across a **surface area** perpendicular to the source, resulting from a **temperature gradient.**

$$Q \propto A \times \frac{dT}{dx} = kA \times \frac{dT}{dx}$$







$$q = k\,A\,\frac{T_1 - T_2}{L}$$

Thermal conductivity [κ, k or λ measured in W/(m·K)] expresses its ability to conduct heat, a rate measurement of **heat flow**, through a material of a **given thickness**, across a **surface area** perpendicular to the source, resulting from a **temperature gradient.**

34.5. Heat Transfer by Conduction through a Slab

Consider a solid slab having one of its face (say left) at a higher temperature and the other (say right) at a lower temperature as shown in Fig. 34.1.

Let

T_1 = Temperature of the left face (*i.e.* higher temperature) in K,

T_2 = Temperature of the right face (*i.e.* lower temperature) in K,

x = Thickness of the slab,

A = Area of the slab.

k = Thermal conductivity of the body,

t = Time through which the heat flow has taken place.

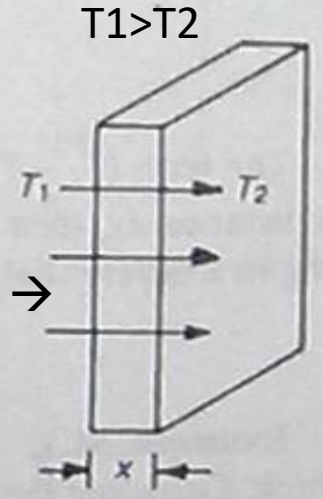


Fig. 34.1. Heat transfer through a slab.

As per the Fourier's law of heat conduction, the heat flow (assuming no loss of heat from the sides) through the slab,

$$Q = kA \times \frac{dT}{dx} = \frac{kA(T_1 - T_2)}{dx}$$

Now the total amount of heat flow in time t may be found out by the equation,

$$Q = \frac{kA(T_1 - T_2)t}{x}$$

Notes : 1. Since the temperature of the slab decreases as x increases, therefore sometimes negative sign is put on the right hand side of the above equation.

2. The rate of heat flow per second is given by the relation,

$$Q = \frac{kA(T_1 - T_2)}{x}$$

In the above equation, if we substitute $A = 1\ m^2$; $(T_1 - T_2) = 1\ K$; $t = 1\ s$ and $x = 1\ m$, then $Q = k$.

It is thus obvious, that the thermal conductivity of a material is numerically equal to the quantity of heat (in joules) which flows in one second through a slab of the material of area $1\ m^2$ and thickness 1 m when its faces differ in temperature by 1 K. It may also be defined as the quantity of heat in joules that flows in one second through one metre cube of a material when opposite faces are maintained at a temperature difference of 1 K.

Notes : 1. The unit of thermal conductivity depends upon the units of the quantities on the right side of the above equation. In S I. units, the unit of thermal conductivity is J/m/K/s = W/m K (1 J/s = 1 W).

2. The above expression for rate of heat flow may also be written as :

$$Q = \frac{k A (T_1 - T_2) t}{x} \qquad Q = \frac{T_1 - T_2}{x / k A}$$

The term $x / k A$ is known as *thermal resistance* and corresponds to resistance 'R' in Ohm's law

34.7. Temperature Gradient

We know that the rate of heat flow through a body,

$$Q = \frac{k A (T_1 - T_2)}{x}$$

The term $(T_1 - T_2)/x$ is known as *temperature gradient*. If the temperature falls by dT over a small distance dx, then temperature gradient may be written as dT/dx. Hence the quantity of heat flowing in a differential form may be written as :

$$Q = k A \times \frac{dT}{dx}$$

34.11. Overall Coefficient of Heat Transfer

In the previous articles, we have discussed only the theoretical cases of heat transfer through conduction. But in actual practice, the heat from a hot body is transferred to the cold body by the combined effect of conduction and convection.

Consider a wall through which heat is transferred from a hot surface to a cold surface as shown in Fig. 34.6. Let

$(T_1 - T_2)$ = Difference of temperatures,

A = Surface area of the wall,

x = Thickness of the wall, and

k = Thermal conductivity of the wall material.

As a matter of fact, there will be a thin film of air on both the hot as well as cold faces of the wall, which will act as transition layers adjacent to the wall surface, and through which the heat also has to flow in addition to the wall as shown in Fig. 34.6. Let A and B be the thickness of the effective film of air for the heat flow.

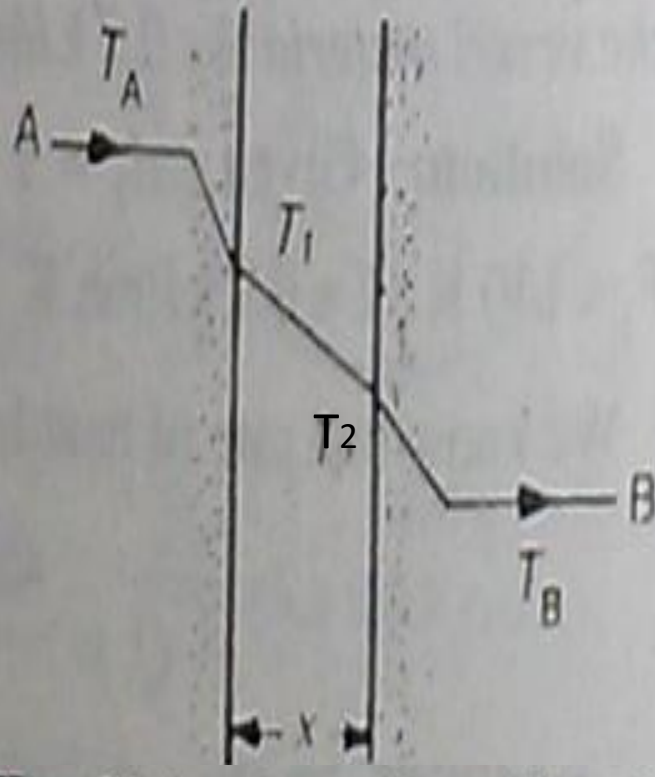


Fig. 34.6 Overall coefficient of heat transfer

T_A and T_B = Temperatures at ends of two thin films of air A and B respectively

h_A and h_B = Coefficients of heat transfer for A and B respectively

U = Overall coefficient of heat transfer

We know that the rate of heat flow through air film A,

$$Q = h_A A (T_A - T_1) \text{ or } (T_A - T_1) = \frac{Q}{h_A A} \quad \ldots (i)$$

Similarly, rate of heat flow through the wall,

$$Q = \frac{kA (T_1 - T_2)}{x} \text{ or } (T_1 - T_2) = \frac{Q x}{k A} \quad \ldots (ii)$$

and rate of heat flow through the film B,

$$Q = h_B A (T_2 - T_B) \text{ or } (T_2 - T_B) = \frac{Q}{h_B A} \quad \ldots (iii)$$

Adding equations (i), (ii) and (iii), we get

$$(T_A - T_B) = \frac{Q}{A}\left[\frac{1}{h_A} + \frac{x}{k} + \frac{1}{h_B}\right]$$

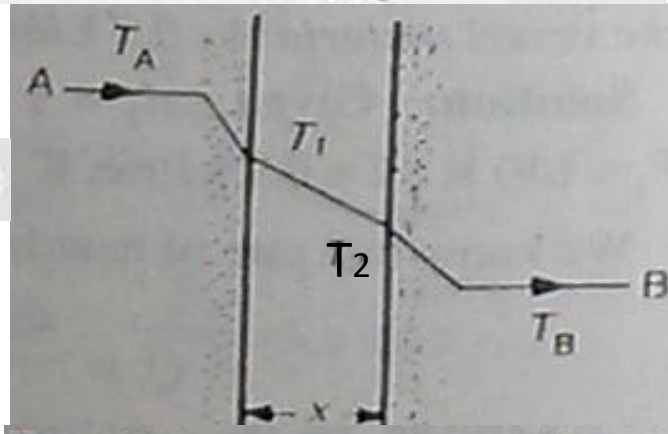


Fig. 34.6 Overall coefficient of heat transfer

$$(T_A - T_B) = \frac{Q}{A}\left[\frac{1}{h_A} + \frac{x}{k} + \frac{1}{h_B}\right]$$

We know that the rate of heat flow,

$$Q = UA\,(T_A - T_B) \quad \ldots (v)$$

$$Q = \frac{A\,(T_A - T_B)}{\left(\frac{1}{h_A} + \frac{x}{k} + \frac{1}{h_B}\right)} \quad \ldots (iv)$$

Now equating the equations (*iv*) and (*v*),

$$U\,A\,(T_A - T_B) = \frac{A\,(T_A - T_B)}{\left(\frac{1}{h_A} + \frac{x}{k} + \frac{1}{h_B}\right)}$$

Overall coefficient of heat transfer

$$U = \frac{1}{\left(\frac{1}{h_A} + \frac{x}{k} + \frac{1}{h_B}\right)}$$

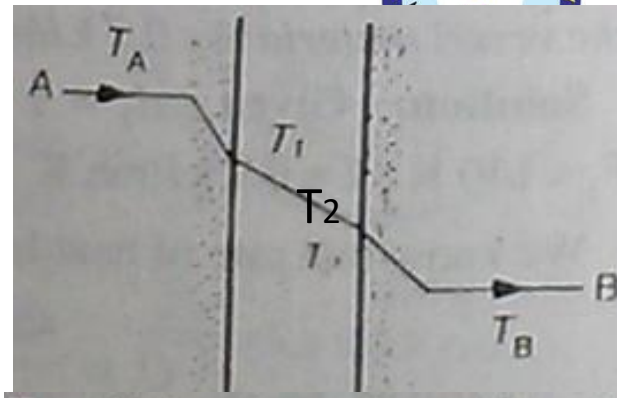


Fig. 34.6 Overall coefficient of heat transfer

$$\left[\frac{2.3}{k}\log\left(\frac{r_2}{r_1}\right)+\frac{1}{r_2 h}\right]$$

$$= \frac{2\pi \times 1\,(452-298)}{\left[\frac{2.3}{0.116}\log\left(\frac{0.16}{0.10}\right)+\frac{1}{0.16\times 9.3}\right]}\ \text{J/s}$$

$$= \frac{967.74}{4.05+0.672} = 205\ \text{J/s}$$ **Ans.**

Note : In this example, thickness and thermal conductivity of pipe as well as heat transfer coefficient of steam is not given. So the example has to be solved ignoring these values.

EXERCISES

1. The temperatures inside and outside of a room are 5° C and 18° C respectively. Calculate the rate at which heat is being transferred through a glass window of area 1 m^2 and thickness 5 mm. Take k for glass as 0.84 W/mK. [Ans. 2.18 kJ/s]

2. The glass windows of a room have total area of 10 square metres, while thickness is 4 mm. Find he rate at which the heat escapes from the room per second by conduction when the temperature difference is 0° C. Take k for glass as 3 W/mK. [Ans. 41.67 kJ/s]

3. An iron boiler 12.5 mm thick contains water at atmospheric pressure. The heated surface is 2 m^2 in rea and the temperature of the inner side is 120° C. If the thermal conductivity of iron is 84 W/mK and the latent eat of evaporation of water is 22.5 J, find the mass of water evaporated. [Ans. 370 kg]

4. An aluminium plate (k = 193.2) and a gold plate (k = 294) are joined with faces in contact. The ter face of the aluminium plate is kept at a temperature of 100° C and the gold face is at 60° C. Compare the ickness required, if the temperature drop through aluminium plate is three times that of gold plate. The surface eas of the two plates is same. [Ans. 197 : 1]

5. A tube has internal radius 20 mm and external radius 25 mm. The inside of the tube is maintained at 20° C. Calculate the quantity of heat conducted through unit length of the tube per [Ans. 8.6 kJ]

Equating the equations (*iv*) and (*v*),

$$U A (T_A - T_B) = \frac{A (T_A - T_B)}{\left(\frac{1}{h_A} + \frac{x}{k} + \frac{1}{h_B} \right)} \quad \ldots (v)$$

$$\therefore \quad U = \frac{1}{\left(\frac{1}{h_A} + \frac{x}{k} + \frac{1}{h_B} \right)}$$

Notes : 1. For the sake of simplicity, we have considered a simple wall of one material only. But this relation may be extended to all the cases discussed earlier, *e.g.* for a composite wall,

$$U = \frac{1}{\left(\frac{1}{h_A} + \frac{x_1}{k_1} + \frac{x_2}{k_2} + ... + \frac{1}{h_B} \right)}$$

Similarly, for a simple cylinder,

$$U = \frac{1}{\left\{ \frac{1}{r_1 h_A} + 2.3 \log \left(\frac{r_2}{r_1} \right) + \frac{1}{r_2 h_B} \right\}}$$

2. If ambient temperature (*i.e.* atmospheric temperature) is existing on one side of the wall or cylinder, then thickness of air film is zero on that side.

Example 34.11. *A furnace wall 200 mm thick is made of a material having thermal conductivity of 1.45 W/m² K. The inner and outer surfaces are exposed to average temperatures of 350° C and 40° C respectively. If the gas and air film coefficients are 58 and 11.63 W/m² K respectively, find the rate of heat transfer through a wall of 2.5 square metre. Also, find the temperatures on the two sides of the wall.*

Solution. Given : $x = 200$ mm $= 0.2$ m ; $k = 1.45$ W/m^2 K ; $T_A = 350°$ C $= 623$ K ; $T_B = 4°$ C $= 313$ K ; $h_A = 58$ W/m^2 K ; $h_B = 11.63$ W/m^2 K ; $A = 2.5$ m^2

Rate of heat transfer

We know that rate of heat transfer,

$$Q = \frac{A\,(T_A - T_B)}{\frac{1}{h_A} + \frac{x}{k} + \frac{1}{h_B}} = \frac{2.5\,(623 - 313)}{\frac{1}{58} + \frac{0.2}{1.45} + \frac{1}{11.63}} = \frac{775}{0.017 + 0.138 + 0.086}\ \text{J/s}$$

$$= 3216\ \text{J/s}\ \textbf{Ans.}$$

Temperature on two sides of the wall

Let T_1 = Temperature on the inner side of the wall, and

T_2 = Temperature on the outside of the wall.

We know that the rate of heat transfer through the gas film, wall and air film is the same. So we shall first consider the flow of heat through the gas film and then through the air film.

We know that $Q = h_A A\,(T_A - T_1)$

$$\therefore \quad 3216 = 58 \times 2.5\,(623 - T_1) = 145\,(623 - T_1)$$

$$623 - T_1 = 3216 / 145 = 22.2 \quad \text{or} \quad T_1 = 600.8\ \text{K} = 327.8°\ \text{C}\ \textbf{Ans.}$$

Similarly $Q = h_B A\,(T_2 - T_B)$

$$3216 = 11.63 \times 2.5\,(T_2 - 313) = 29\,(T_2 - 313)$$

$$\therefore \quad T_2 = \frac{3216}{29} + 313 = 423.9\ \text{K} = 150.9°\ \text{C}\ \textbf{Ans.}$$

Example 34.12. *A composite wall is made up of brickwork, fiberglass and insulating board with thickness and thermal conductivities as given below :*

Brick work 110 mm thick 1.15 W/mK